Lin Ma
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Personal Statement

I am fascinated by the challenge of transforming unstructured social media data into structured business intelligence. My experience in text processing and data visualization has shown me the power of sentiment analysis and trend mapping. This project represents the ideal next step to apply and extend these skills within a rigorous business context, moving beyond technical analysis to deliver actionable recommendations that can directly impact marketing or product strategy.

I am drawn to the practical challenge of creating an interactive dashboard that synthesizes historical data, rolling forecasts, and scenario models. This project excites me as it requires not only technical skills in forecasting and dashboard development but also a deep engagement with business processes to ensure the tool delivers true financial visibility and aligns tactical tender activity with long-term revenue goals.

I am motivated by the goal of creating transparent and explainable analytical systems. In an area as critical as grant allocation, I believe analytics should empower human decision-makers with clear, auditable logic rather than act as a black box. This project offers a meaningful opportunity to combine financial analysis with principles of explainable AI and clear communication, ensuring that the tool's assessments are both robust and understandable to its users.

Summary

- Software Development & Engineering Excellence: Designed and implemented a full-stack web application using React and Node.js, incorporating secure user authentication to enable robust and scalable interaction.

- Advanced Programming & Algorithmic Problem-Solving: Proficient in Python and C++, with a solid foundation in data structures and algorithms. Capable of designing efficient solutions to complex computational problems.

- Data Engineering & System Architecture: Architected a scalable data pipeline for a semester project that efficiently ingested, processed, and stored real-time sensor data using Python and SQL, demonstrating strong system design principles.

- Cross-functional Team Leadership & Agile Collaboration: Led development within a team using GitHub for version control, facilitating sprint planning and enabling the delivery of all core features ahead of schedule.

- Full-Stack Development & System Integration: Engineered a complete data visualization solution by seamlessly integrating a Python Flask backend with a React.js frontend and a MySQL database, enabling dashboarding of key indicators.

- Statistical Analysis & Modeling with R: Conducted comprehensive data analysis and built predictive statistical models using R Studio (tidyverse, caret) for academic research, effectively interpreting results and generating publication-ready visualizations with ggplot2.

- Data Reporting & Automation: Developed reproducible data analysis reports and automated weekly business metric dashboards using R Markdown in R Studio, streamlining the insights delivery process for stakeholder review.

Education

Master of Professional Studies in Data Science — 07/2025 - now
University of Auckland — Auckland, New Zealand
Bachelor of Computer Science — 09/2021 – 06/2025
Southwest University — Chongqing, China